



UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Advanced Level

MATHEMATICS

9709/33

Paper 3 Pure Mathematics 3 (P3)

May/June 2011

1 hour 45 minutes

Additional Materials: Answer Booklet/Paper
Graph Paper
List of Formulae (MF9)

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 75.

Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.

This document consists of 3 printed pages and 1 blank page.



- 1 Use logarithms to solve the equation $5^{2x-1} = 2(3^x)$, giving your answer correct to 3 significant figures. [4]

- 2 The curve $y = \frac{\ln x}{x^3}$ has one stationary point. Find the x -coordinate of this point. [4]

- 3 Show that $\int_0^1 (1-x)e^{-\frac{1}{2}x} dx = 4e^{-\frac{1}{2}} - 2$. [5]

- 4 (i) Show that the equation

$$\tan(60^\circ + \theta) + \tan(60^\circ - \theta) = k$$

can be written in the form

$$(2\sqrt{3})(1 + \tan^2 \theta) = k(1 - 3 \tan^2 \theta). \quad [4]$$

- (ii) Hence solve the equation

$$\tan(60^\circ + \theta) + \tan(60^\circ - \theta) = 3\sqrt{3},$$

giving all solutions in the interval $0^\circ \leq \theta \leq 180^\circ$. [3]

- 5 The polynomial $ax^3 + bx^2 + 5x - 2$, where a and b are constants, is denoted by $p(x)$. It is given that $(2x - 1)$ is a factor of $p(x)$ and that when $p(x)$ is divided by $(x - 2)$ the remainder is 12.

- (i) Find the values of a and b . [5]

- (ii) When a and b have these values, find the quadratic factor of $p(x)$. [2]

- 6 (i) By sketching a suitable pair of graphs, show that the equation

$$\cot x = 1 + x^2,$$

where x is in radians, has only one root in the interval $0 < x < \frac{1}{2}\pi$. [2]

- (ii) Verify by calculation that this root lies between 0.5 and 0.8. [2]

- (iii) Use the iterative formula

$$x_{n+1} = \tan^{-1} \left(\frac{1}{1 + x_n^2} \right)$$

to determine this root correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

- 7 (i) Find the roots of the equation

$$z^2 + (2\sqrt{3})z + 4 = 0,$$

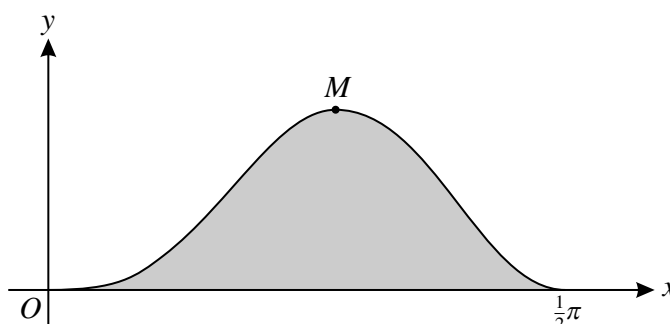
giving your answers in the form $x + iy$, where x and y are real. [2]

- (ii) State the modulus and argument of each root. [3]

- (iii) Showing all your working, verify that each root also satisfies the equation

$$z^6 = -64. \quad [3]$$

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The diagram shows the curve $y = 5 \sin^3 x \cos^2 x$ for $0 \leq x \leq \frac{1}{2}\pi$, and its maximum point M .

- (i) Find the x -coordinate of M . [5]

- (ii) Using the substitution $u = \cos x$, find by integration the area of the shaded region bounded by the curve and the x -axis. [5]

- 9 In a chemical reaction, a compound X is formed from two compounds Y and Z . The masses in grams of X , Y and Z present at time t seconds after the start of the reaction are x , $10 - x$ and $20 - x$ respectively. At any time the rate of formation of X is proportional to the product of the masses of Y and Z present at the time. When $t = 0$, $x = 0$ and $\frac{dx}{dt} = 2$.

- (i) Show that x and t satisfy the differential equation

$$\frac{dx}{dt} = 0.01(10 - x)(20 - x). \quad [1]$$

- (ii) Solve this differential equation and obtain an expression for x in terms of t . [9]

- (iii) State what happens to the value of x when t becomes large. [1]

- 10 With respect to the origin O , the lines l and m have vector equations $\mathbf{r} = 2\mathbf{i} + \mathbf{k} + \lambda(\mathbf{i} - \mathbf{j} + 2\mathbf{k})$ and $\mathbf{r} = 2\mathbf{j} + 6\mathbf{k} + \mu(\mathbf{i} + 2\mathbf{j} - 2\mathbf{k})$ respectively.

- (i) Prove that l and m do not intersect. [4]

- (ii) Calculate the acute angle between the directions of l and m . [3]

- (iii) Find the equation of the plane which is parallel to l and contains m , giving your answer in the form $ax + by + cz = d$. [5]

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